Proprietary to General Electric Company
Direction 5438996-8EN, **Rev#** 9
Brivo CT385 Series Service Methods (Adv)
Module Version 1.0, **Version Date** 3/14/13

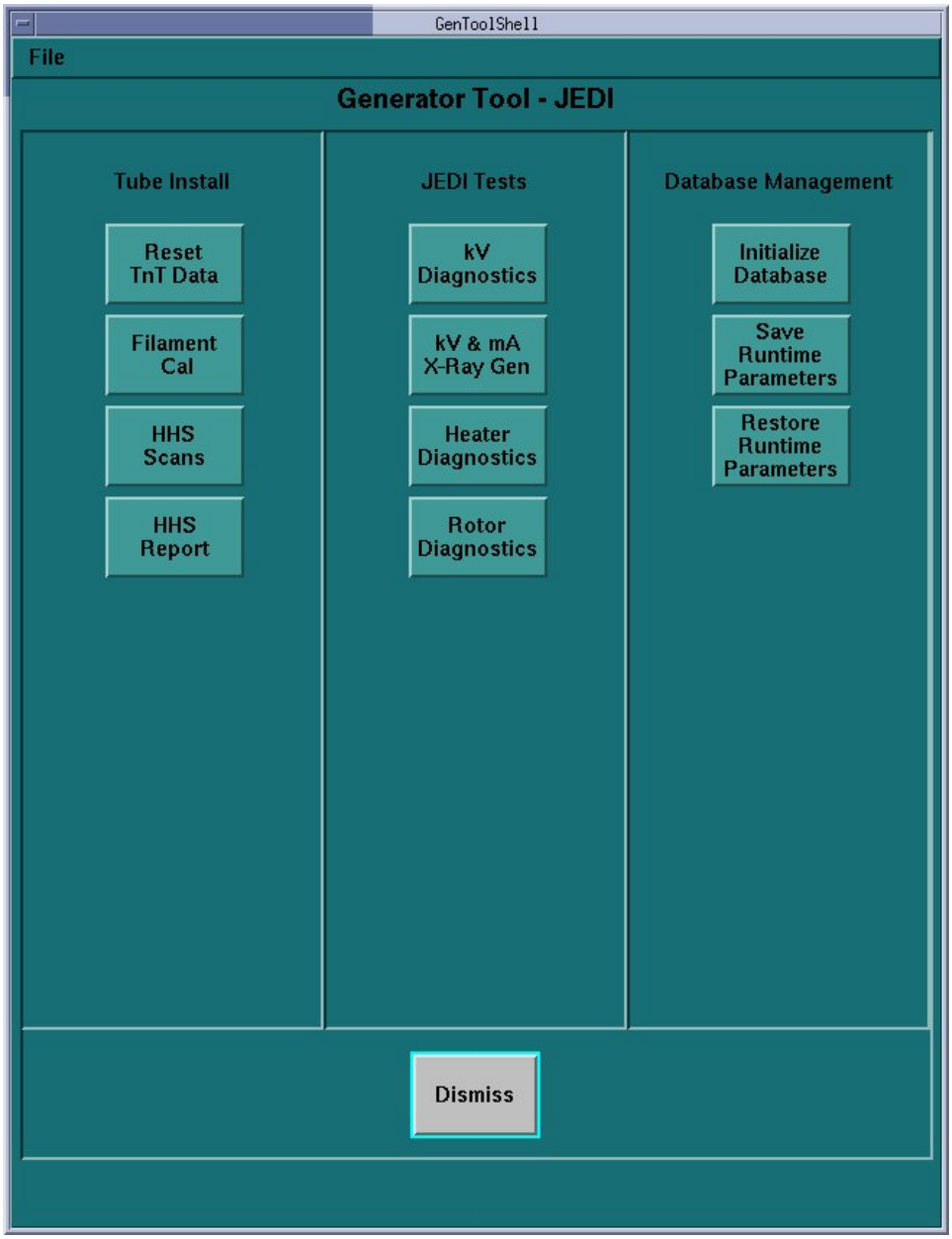
JEDI Generator Tool

Last Revised:	17 May, 2012

1 Introduction

The JEDI Generator Tool is launched from the Common Service Desktop by clicking [DIAGNOSTICS] > [X-RAY GENERATION]. This tool contains many key service features for the X-Ray Generation subsystem.

Illustration 1: JEDI Generator Tool



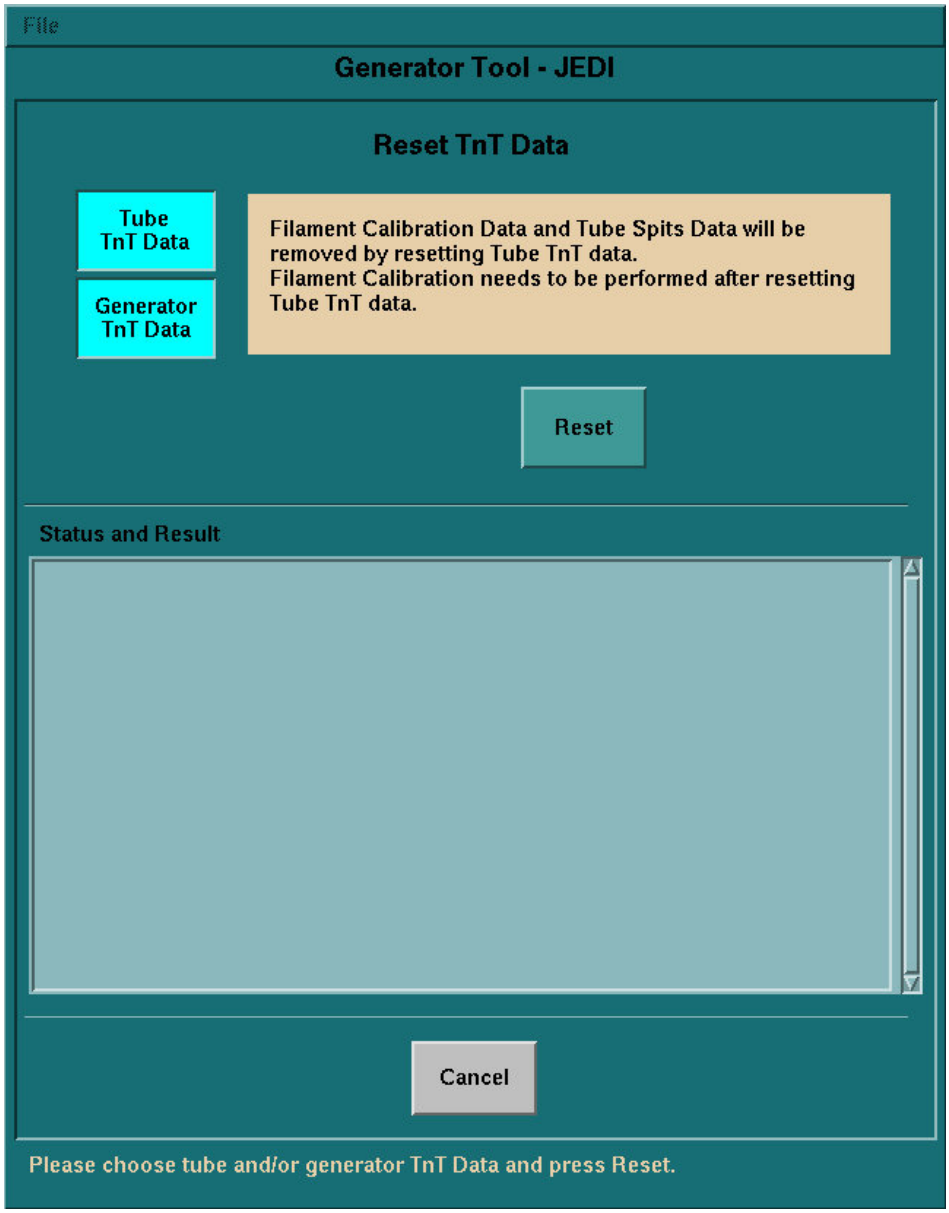
This tool can be thought of as the "portal" to many of the important X-Ray generation features. The tools launched from within Generator Tool - JEDI will consist of 3 main areas: the Message area, Status and Result screen, and Instruction area (see Illustration 2).

2 Tube Install

2.1 Reset TnT Data

Use this tool to reset the Tracking and Trending data stored in the JEDI Generator. This tool can reset either the Tube or the Generator TnT data. To initiate the selected operation, press [Reset].

Illustration 2: Reset TnT Data Tool



In Illustration 2, note the Message Area at the top (tan), the Status and Result Screen in the middle (light green) and the Instruction Area at the bottom (dark green).

2.2 Filament Calibration

The Filament Calibration tool is used to calibrate the JEDI Generator based on the X-Ray Tube. It is part of the Generator Characterization procedure. Pressing [Scan] begins the calibration process. If this program hangs, it does not overwrite the current calibration parameters stored in the JEDI database. The same holds true if the user cancels the process.

This tool performs a pre-determined scan sequence and displays the results in the Status and Result screen. Both the small and large filaments to must pass for the system to scan properly.

In order for Filament Calibration to run properly, the system must be able to scan. As such, the parameters stored in the JEDI generator database must be close to calibrated. Loading the "default database" will always provide a baseline.

2.3 HHS Scans

Use the HHS Scans tool to qualify that the system has properly calibrated the generator subsystem. In many places, law requires this when a high voltage component has been replaced. See the installation manual for more detail on HHS compliance / Data Verification.

The program, when run, launches a scan sequence and takes measurements throughout the sequence to verify that the system is working within specifications.

3 JEDI Tests

NOTE: For systems with certain system software versions, you will encounter the following problem with Jedi Tests:

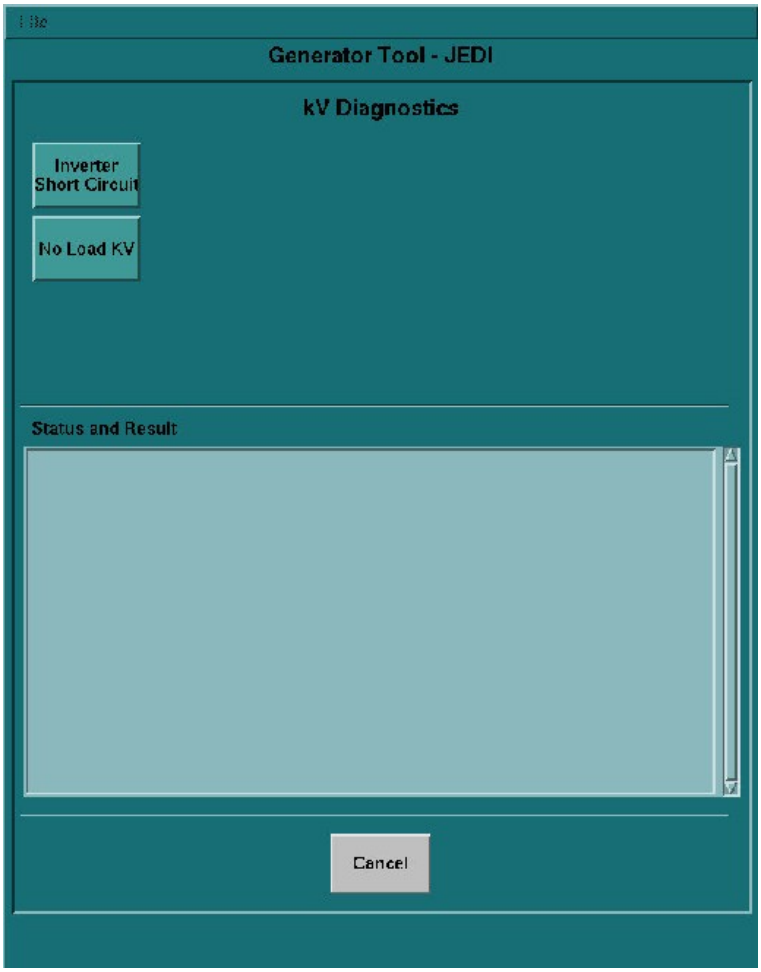
After exiting the kV Diagnostics menu (after executing the diagnostics or clicking [Cancel]), and when you have entered the Heater Diagnostics or Rotor Diagnostics menu, boxes showing the diagnostics procedure and a Start button will not be displayed.

When this happens, exit the Generator Tool and re-enter it. You can then normally execute the Heater Diagnostics or Rotor Diagnostics. This means that when you want to execute the Heater or Rotor Diagnostics, first execute it after entering the Generator Tool.

3.1 kV Diagnostics

Clicking on the kV Diagnostics button will launch a separate screen with three diagnostic tools that are available for troubleshooting the X-Ray Generation subsystem. In order to run these tools, hardware interaction is required.

Illustration 3: kV Diagnostics (Example - actual may vary slightly)



Each tool provides instructions for setting up the test as shown in Illustration 4. Each test asks for confirmation that the necessary steps have been performed. Following that, each tool accepts entry of setup parameters (defaults will be given) before performing the test (see Illustration 5).

Illustration 4: Example of test with instructions

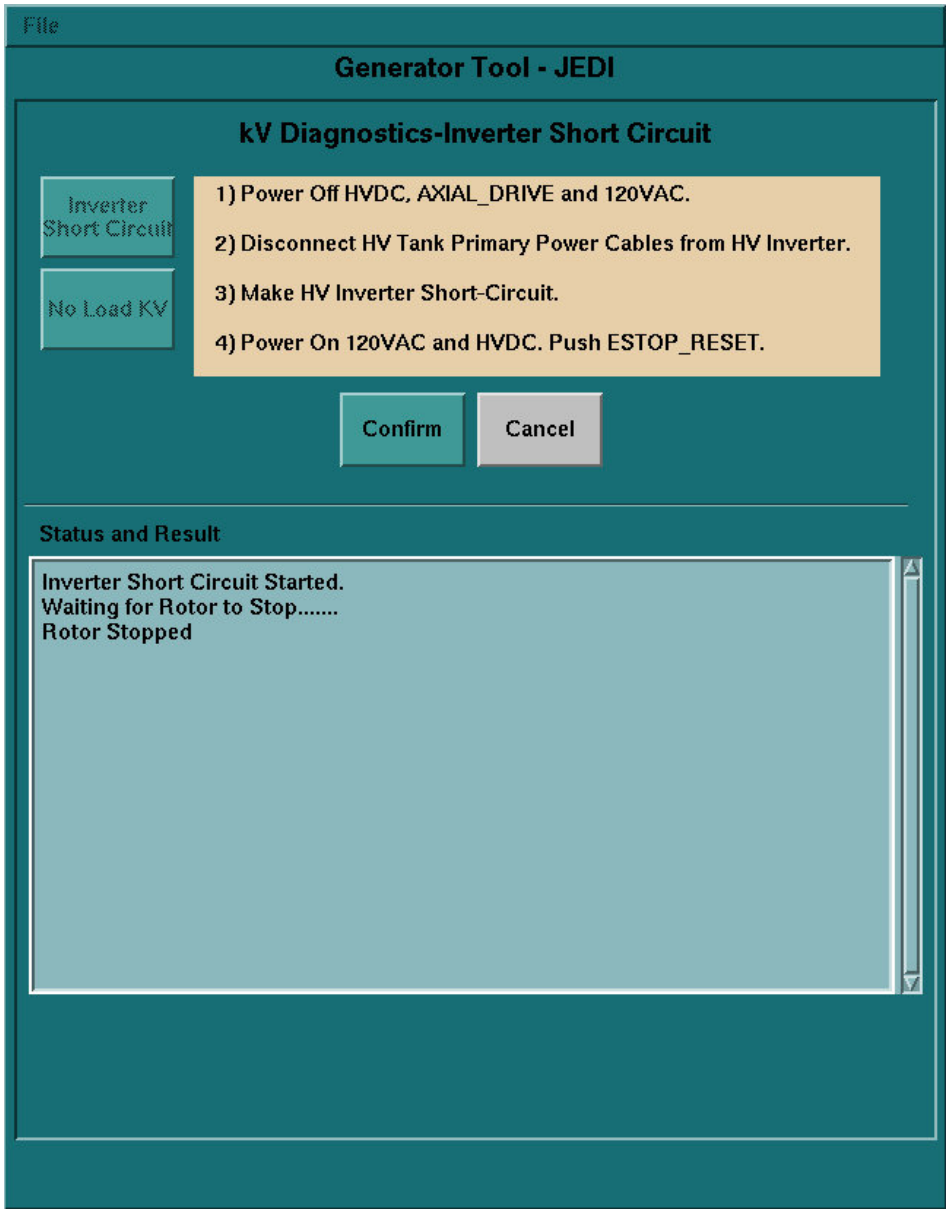
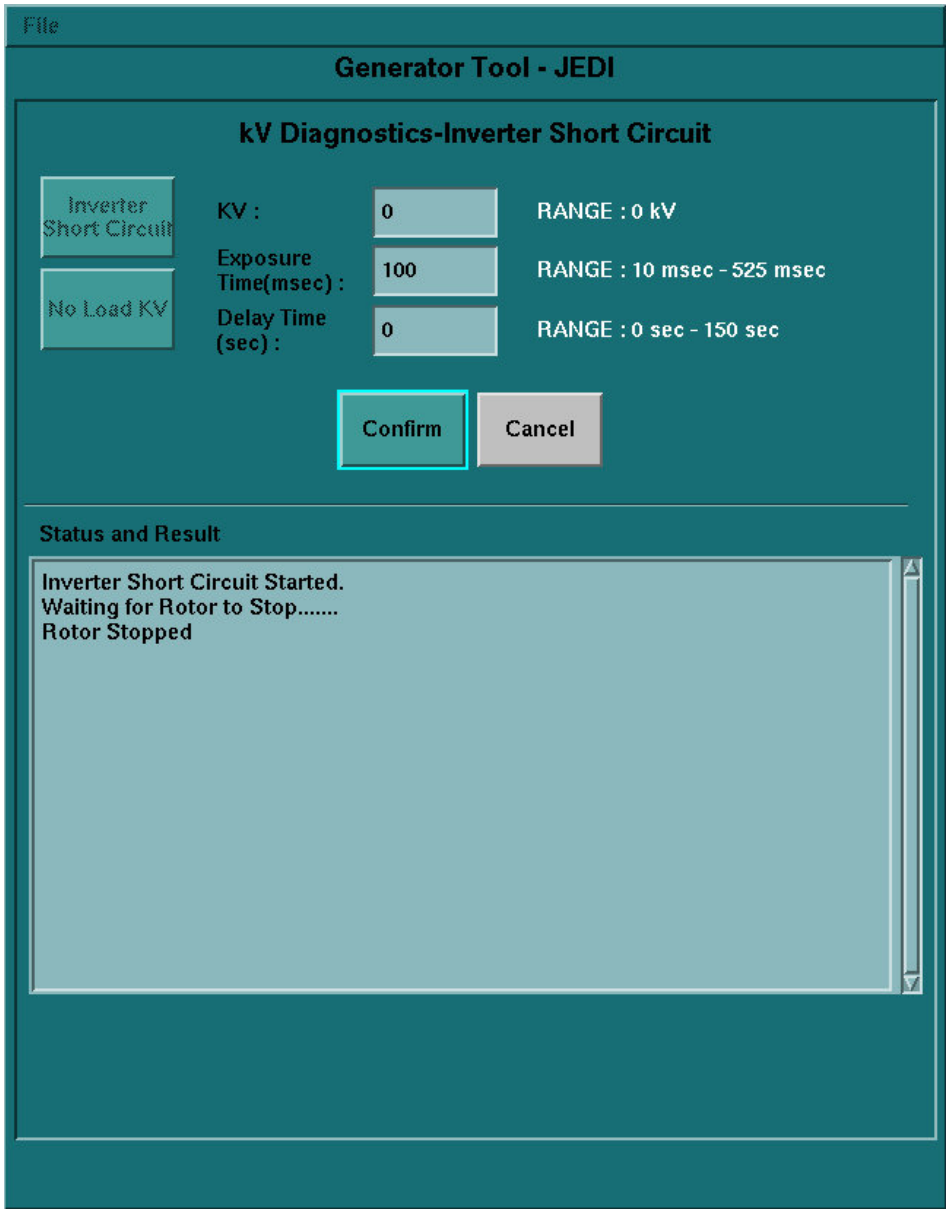


Illustration 5: Example of test and parameters



3.2 kV & mA X-Ray Gen

This tool can be used to scan in a diagnostic-type setting. It allows for collecting High Voltage statistics during a single x-ray exposure. Virtually all possible single exposure scan types at the system level are allowed.Illustration 6 provides a sample screen shot.

The kV & mA (X-Ray) tool can also upload (to a file on the console) or view the TnT data. You must upload the data before you view it. The upload feature overwrites any previous data stored in the file.

Illustration 6: kV & mA (X-Ray) tool

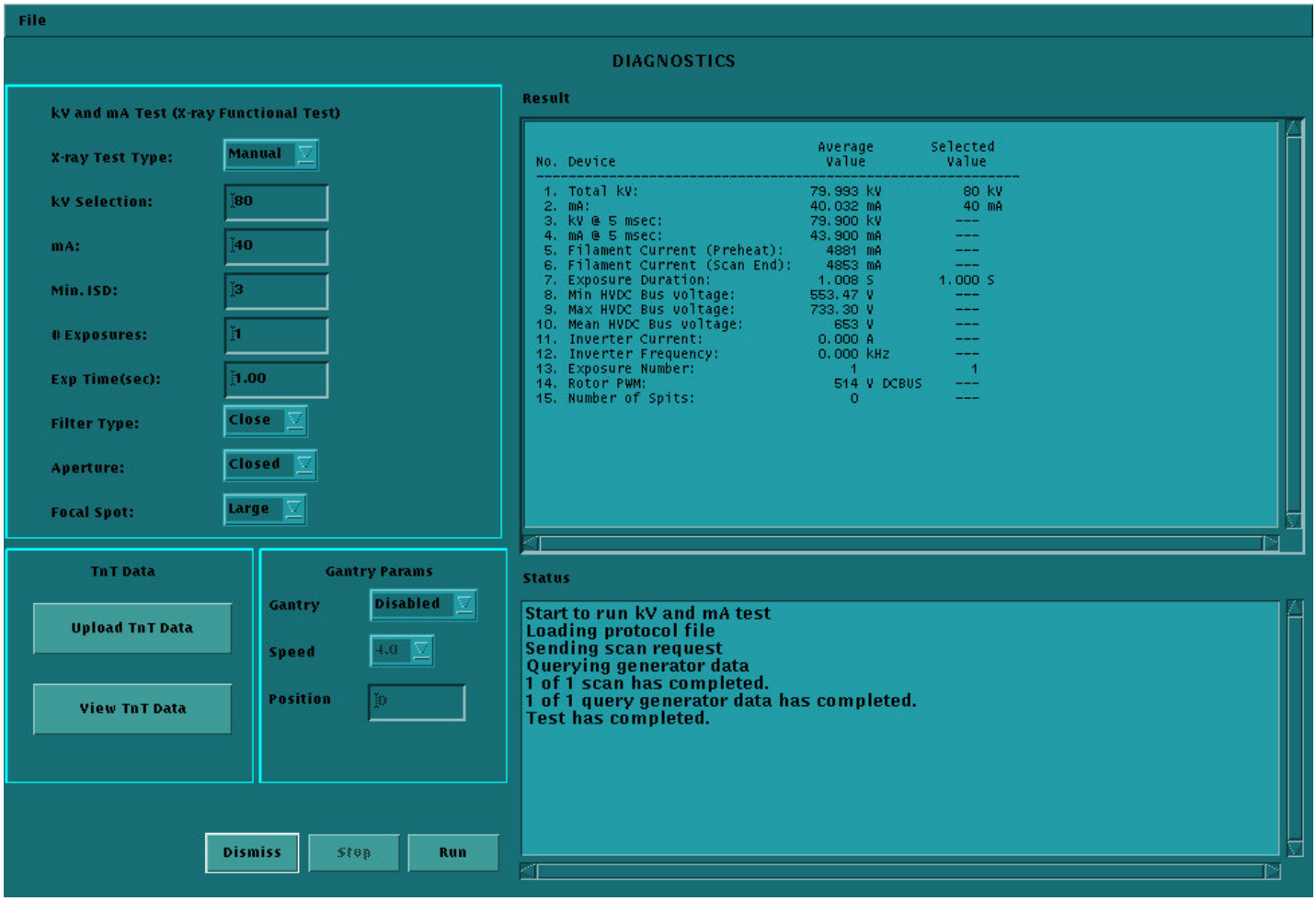


Table 1: kV & mA (X-Ray) Tool Diagnostic Measurements

Measurement	Explanation
*Total kV:	Scan kV
*mA:	Measured mA during scan
*kV @ 5 msec:	Measured kV around the 5 ms (approx. 4-6 sec) point in the scan
*mA @ 5 msec:	Measured mA around the 5 ms (approx. 4-6 sec) point in the scan
Filament Current (Preheat):	Measured filament current during preheat phase
Filament Current (Scan End):	Measured filament current at the end of the scan
Exposure Duration	Measured exposure duration
Min HVDC Bus Voltage	Minimum HVDC Bus Voltage measured during exposure
Max HVDC Bus Voltage	Maximum HVDC Bus Voltage measured during exposure
*Mean HVDC Bus Voltage	Mean of all HVDC measured values
*Inverter Current	Measured inverter current
*Inverter Frequency	Measured inverter frequency
Exposure Number	Exposure number that just finished (will display statistics if user cancels)
Rotor PWM	Recorded Voltage necessary to drive Anode Rotation
Number of Spits	Total count of tube spits during exposure

3.3 Heater Diagnostics

The Heater Diagnostic tool runs a sequence that drives the tube filaments. If an error occurs, the diagnostic notifies of the error. If this occurs, check the system error log.

3.4 Rotor Diagnostics

The Rotor Diagnostic tool runs a sequence that rotates the anode. If an error occurs, the diagnostic notifies of the error. If this occurs, check the system error log.

4 Database Management

The database management tools allow for uploading and / or downloading the following data within the database found on the PPC kV control board.

- Tube TnT
- Generator TnT
- Event Log

[Initialize Database] downloads the default data above to the control board.

[Save Runtime Parameters] uploads the data to a file on the console.

[Restore Runtime Parameters] downloads the saved data (by Save Runtime Parameters) to the control board. If Save Runtime Parameters has not ever been used, the button is shown dimmed.

To download all the default database (including other data than mentioned above) to the JEDI generator, use the Flash Download Tool.

Back to top